

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Subject: 392131 Physics I

Date: 6 October 2014

Year: 2014

Section: 15 - 16

Time: 13.00 – 15.00

Name _____ ID _____ Class _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken inside the examination room.
3. Textbooks and dictionaries are **NOT** allowed.
4. Express your answer in English. Other languages will be discarded.
5. A scientific calculator is allowed.
6. Electronic communication devices are **NOT** allowed in the examination room.
7. The examination has 6 pages (including this page), 15 questions and a total score of 55 points.
8. Write all your answers on this exam sheet.

1. Show that the expression $as = \frac{1}{2}(v^2 - u^2)$ is dimensionally correct. (1 point)

2. Determine the number of significant figures in the following measured values. (3 points)

a) 23 cm	b) 3.589 s	c) $4.67 \times 10^{-3} \text{ m/s}$
d) 0.0032 m	e) $42.44 \pm 0.01 \text{ g}$	

3. Use the rules for significant figures to find the answer to the addition problem. (1 point)
 $14.4 + 0.234 + 108.17 + 4.567 =$

4. A carpet is to be installed in a room whose length is measured to be 12.71 m and whose width is measured to be 3.46 m. Find the area of the room. (2 points)

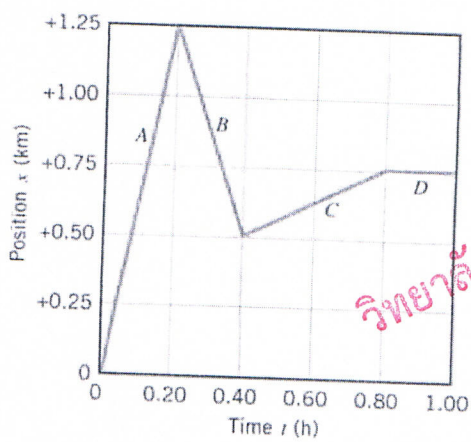
5. The volume of a wallet is 8.50 in^3 . Convert this value to m^3 . (1 in. = 0.0254 m = 2.54 cm) (2 points)

6. Assume that it takes 7.00 minutes to fill a 30.0-gal gasoline tank. (4 points)
 - (a) Calculate the rate at which the tank is filled in gallons per second.
 - (b) Calculate the rate at which the tank is filled in cubic meters per second.
 - (c) Determine the time interval, in hours, required to fill a 1 m^3 volume at the same rate.

(1 gal = 231 in.³ and 1 in. = 0.0254 m)

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7. Two points are given in polar coordinates by $(x,y) = (5.00 \text{ m}, -3 \text{ m})$ and, $(r, \theta) = (7.00 \text{ m}, 280.0^\circ)$ respectively. What is the distance between them? (5 points)



8. A person who walks for exercise produces the position-time graph given with this problem. Calculate the average velocity for each segment. (4 points)

9. A particle moves according to the equation $x = 10t^2$ where x is in meters and t is in seconds. Find the average velocity for the time interval from 3.00s to 6.00s. (3 points)
10. In getting ready to slam-dunk the ball, a basketball player starts from rest and sprints to a speed of 6.0 m/s in 1.5 s. Assuming that the player accelerates uniformly, determine the distance he runs. (2 points)
11. A speedboat starts from rest and accelerates at $+2.01 \text{ m/s}^2$ for 7.00 s. At the end of this time, the boat continues for an additional 6.00 s with an acceleration of $+0.518 \text{ m/s}^2$. Following this, the boat accelerates at -1.49 m/s^2 for 8.00 s. |
- a) What is the velocity of the boat at $t = 21.0 \text{ s}$?
- b) Find the total displacement of the boat. (7 points)

12. A jogger accelerates from rest to 3.0 m/s in 2.0 s . A car accelerates from 38.0 to 41.0 m/s also in 2.0 s .
- Find the acceleration of the jogger.
 - Determine the acceleration of the car.
 - Does the car travel farther than the jogger during the 2.0 s ? If so, how much farther? (6 points)

13. A ball is thrown vertically upward with a speed of 50 m/s . (4 points)

- How high does it rise?
- How long does it take to reach its highest point?

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14. A model rocket blasts off from the ground, rising straight upward with a constant acceleration that has a magnitude of 86.0 m/s^2 for 1.70 seconds, at which point its fuel abruptly runs out. Air resistance has no effect on its flight. What maximum altitude (above the ground) will the rocket reach? (6 points)

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15. A cement block accidentally falls from rest from the ledge of a 53.0 m high building. When the block is 14.0 m above the ground, a man, 2.00 m tall, looks up and notices that the block is directly above him. How much time, at most, does the man have to get out of the way? (5 points)